

## GROUP D MEMEBERS

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## LAB 1 – VOIP CALLS INTERCERPTION AND PACKETS ANALYSIS

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### Aims

The aim of this lab is to understand and demonstrate the process of VoIP traffic interception and analysis within a controlled virtual environment using Linphone and Wireshark.

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### Objectives

1. To configure and successfully initiate a VoIP call using Linphone.
  2. To capture VoIP network traffic during an active call using Wireshark.
  3. To identify VoIP-related protocols such as SIP, RTP, and RTCP in the captured traffic.
  4. To analyze detailed packet information including source IP, destination IP, ports, and call setup messages.
  5. To understand how VoIP communication can be monitored in a test lab environment.
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### Methodology

- We install the virtual box machine on the pc
- We install the kali linux operating system inside virtual box machine

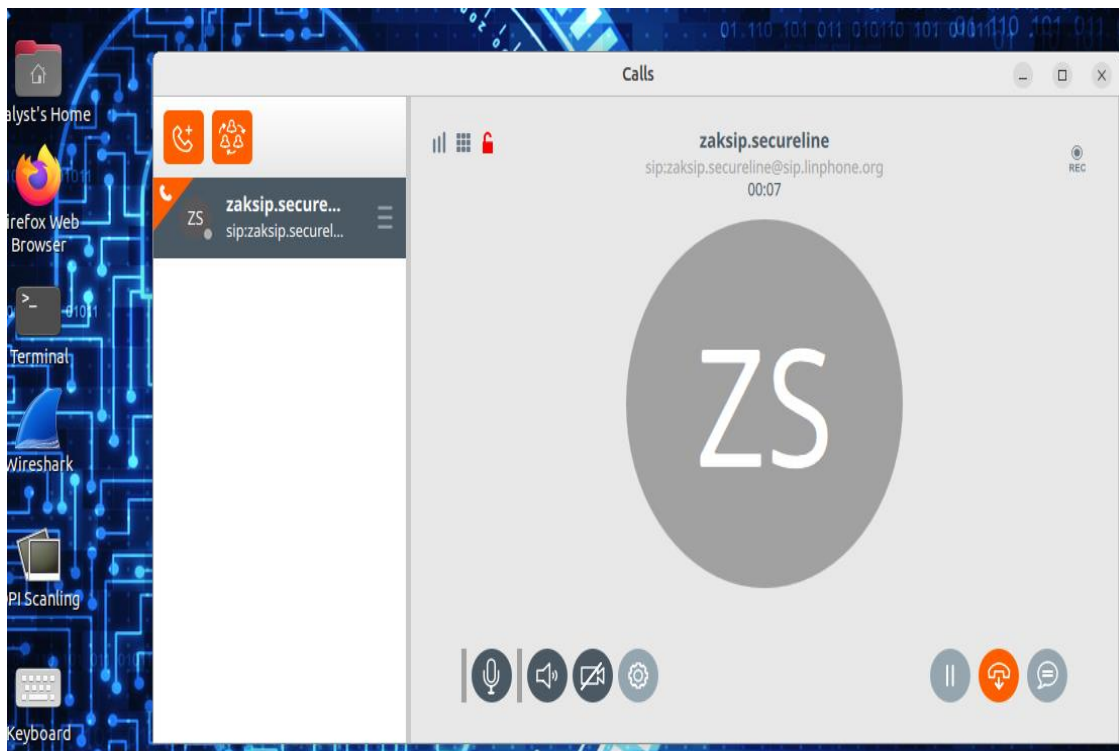
- We clone the kali linux operating system 2 times to have 3 kali linux operating system
- We configure all the 3 kali linux OS to NAT network.
- We powered on the 3 kali linux operating system
- We install linphone (soft phone) application inside 2 of the kali linux OS using the command **sudo apt install linphone**
- We open the linphone application that was installed inside 2 of the clone kali linux OS
- We initiate a call from one of the clone kali linux OS using the ip address of the second clone kali linux OS
- We open the wireshark application on the third clone kali linux OS to capture the traffic between the 2 clone kali
- We saved the captured traffic as voip.pcap.

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## Screenshot

### SCREENSHOWING SHOWING SUCCESSFULLY CALL INITIATION USING LINPHONE





## SCREENSHOT SHOWING SUCCESSFUL CAPTURING OF VOIP TRAFFIC USING WIRESHARK

| *enp0s3                                                                                          |             |          |              |          |        |                                                                     |
|--------------------------------------------------------------------------------------------------|-------------|----------|--------------|----------|--------|---------------------------------------------------------------------|
| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help                       |             |          |              |          |        |                                                                     |
| Apply a display filter ... <Ctrl-/>                                                              |             |          |              |          |        |                                                                     |
| No.                                                                                              | Time        | Source   | Destination  | Protocol | Length | Info                                                                |
| 104                                                                                              | 1.964333035 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52                                                 |
| 105                                                                                              | 1.973384528 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52                                                 |
| 106                                                                                              | 1.994686963 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52                                                 |
| 107                                                                                              | 2.004747914 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52                                                 |
| 108                                                                                              | 2.033797997 | 10.0.2.6 | 5.135.215.43 | IPv4     | 1514   | Fragmented IP protocol (proto=UDP 17, off=0, ID=5d98) [Reassembl... |
| 109                                                                                              | 2.033825313 | 10.0.2.6 | 5.135.215.43 | SIP/SDP  | 138    | Status: 200 Ok (INVITE)                                             |
| 110                                                                                              | 2.043656999 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=272, Time=2319627665                  |
| 111                                                                                              | 2.054441657 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=273, Time=2319628625                  |
| 112                                                                                              | 2.074486480 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=274, Time=2319629585                  |
| 113                                                                                              | 2.094346672 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=275, Time=2319630545                  |
| 114                                                                                              | 2.123507288 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=276, Time=2319631505                  |
| 115                                                                                              | 2.133855369 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=277, Time=2319632465                  |
| 116                                                                                              | 2.155230069 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=278, Time=2319633425                  |
| 117                                                                                              | 2.173981388 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=279, Time=2319634385                  |
| 118                                                                                              | 2.183964111 | 10.0.2.6 | 5.135.215.43 | STUN     | 62     | Binding Request                                                     |
| 119                                                                                              | 2.205045311 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=280, Time=2319635345                  |
| 120                                                                                              | 2.214334402 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=281, Time=2319636305                  |
| 121                                                                                              | 2.233860798 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=282, Time=2319637265                  |
| 122                                                                                              | 2.243983825 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=283, Time=2319638225                  |
| 123                                                                                              | 2.286466295 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=284, Time=2319639185                  |
| 124                                                                                              | 2.303421255 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=285, Time=2319640145                  |
| 125                                                                                              | 2.313635438 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=286, Time=2319641105                  |
| ▶ Frame 125: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface enp0s3, id 0 |             |          |              |          |        |                                                                     |

|     |             |          |              |      |     |                                                    |
|-----|-------------|----------|--------------|------|-----|----------------------------------------------------|
| 191 | 3.573322546 | 10.0.2.6 | 5.135.215.43 | RTCP | 320 | Sender Report                                      |
| 192 | 3.583672871 | 10.0.2.6 | 5.135.215.43 | SRTP | 94  | PT=opus, SSRC=0x3382BC34, Seq=349, Time=2319701585 |
| 193 | 3.593688246 | 10.0.2.6 | 5.135.215.43 | SRTP | 94  | PT=opus, SSRC=0x3382BC34, Seq=350, Time=2319702545 |
| 194 | 3.603531729 | 10.0.2.6 | 5.135.215.43 | SRTP | 94  | PT=opus, SSRC=0x3382BC34, Seq=351, Time=2319703505 |

## SCREENSHOT SHOWING DETAIL INFORMATION ABOUT THE VOIP TRAFFIC CAPTURED

\*enp0s3

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Apply a display filter ... <Ctrl-/>

| No. | Time        | Source   | Destination  | Protocol | Length | Info                |
|-----|-------------|----------|--------------|----------|--------|---------------------|
| 104 | 1.964333035 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52 |
| 105 | 1.973384528 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52 |
| 106 | 1.994686963 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52 |

▶ Frame 105: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface enp0s3, id 0

▼ Ethernet II, Src: PcsCompu 55:44:07 (08:00:27:55:44:07), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)

- ▶ Destination: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
- ▶ Source: PcsCompu 55:44:07 (08:00:27:55:44:07)
- Type: IPv4 (0x0800)

▼ Internet Protocol Version 4, Src: 10.0.2.6, Dst: 5.135.215.43

- 0100 .... = Version: 4
- .... 0101 = Header Length: 20 bytes (5)
- ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 80
- Identification: 0x5d90 (23952)
- Flags: 0x40, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: UDP (17)
- Header Checksum: 0xf454 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.0.2.6
- Destination Address: 5.135.215.43

▶ User Datagram Protocol, Src Port: 7078, Dst Port: 12134

\*enp0s3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

| No. | Time        | Source   | Destination  | Protocol | Length | Info                                                                |
|-----|-------------|----------|--------------|----------|--------|---------------------------------------------------------------------|
| 107 | 2.004747914 | 10.0.2.6 | 5.135.215.43 | UDP      | 94     | 7078 → 12134 Len=52                                                 |
| 108 | 2.033797997 | 10.0.2.6 | 5.135.215.43 | IPv4     | 1514   | Fragmented IP protocol (proto=UDP 17, off=0, ID=5d98) [Reassembl... |

▶ Frame 108: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface enp0s3, id 0

▼ Ethernet II, Src: PcsCompu 55:44:07 (08:00:27:55:44:07), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)

- ▶ Destination: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
- ▶ Source: PcsCompu 55:44:07 (08:00:27:55:44:07)
- Type: IPv4 (0x0800)

▼ Internet Protocol Version 4, Src: 10.0.2.6, Dst: 5.135.215.43

- 0100 .... = Version: 4
- .... 0101 = Header Length: 20 bytes (5)
- ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 1500
- Identification: 0x5d98 (23960)
- Flags: 0x20, More fragments
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: UDP (17)
- Header Checksum: 0x0ec1 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.0.2.6
- Destination Address: 5.135.215.43
- [\[Reassembled IPv4 in frame: 1091\]](#)

▼ Data (1400 bytes)

\*enp0s3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

| No. | Time        | Source   | Destination  | Protocol | Length | Info                    |
|-----|-------------|----------|--------------|----------|--------|-------------------------|
| 109 | 2.033825313 | 10.0.2.6 | 5.135.215.43 | SIP/SDP  | 138    | Status: 200 Ok (INVITE) |

- Frame 109: 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface enp0s3, id 0
- ▾ Ethernet II, Src: PcsCompu 55:44:07 (08:00:27:55:44:07), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
  - Destination: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
  - Source: PcsCompu 55:44:07 (08:00:27:55:44:07)
  - Type: IPv4 (0x0800)
- ▾ Internet Protocol Version 4, Src: 10.0.2.6, Dst: 5.135.215.43
  - 0100 .... = Version: 4
  - .... 0101 = Header Length: 20 bytes (5)
  - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  - Total Length: 124
  - Identification: 0x5d98 (23960)
  - Flags: 0x00
  - ...0 0101 1100 1000 = Fragment Offset: 1480
  - Time to Live: 64
  - Protocol: UDP (17)
  - Header Checksum: 0x3368 [validation disabled]
  - [Header checksum status: Unverified]
  - Source Address: 10.0.2.6
  - Destination Address: 5.135.215.43
  - [2 IPv4 Fragments (1584 bytes): #108(1480), #109(104)]
- User Datagram Protocol, Src Port: 5060, Dst Port: 5060
- Session Initiation Protocol (200)

\*enp0s3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

| No. | Time        | Source   | Destination  | Protocol | Length | Info                                               |
|-----|-------------|----------|--------------|----------|--------|----------------------------------------------------|
| 110 | 2.043656999 | 10.0.2.6 | 5.135.215.43 | SRTP     | 94     | PT=opus, SSRC=0x3382BC34, Seq=272, Time=2319627665 |

- Frame 110: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface enp0s3, id 0
- ▾ Ethernet II, Src: PcsCompu 55:44:07 (08:00:27:55:44:07), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
  - Destination: 52:55:0a:00:02:01 (52:55:0a:00:02:01)
  - Source: PcsCompu 55:44:07 (08:00:27:55:44:07)
  - Type: IPv4 (0x0800)
- ▾ Internet Protocol Version 4, Src: 10.0.2.6, Dst: 5.135.215.43
  - 0100 .... = Version: 4
  - .... 0101 = Header Length: 20 bytes (5)
  - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  - Total Length: 80
  - Identification: 0x5d99 (23961)
  - Flags: 0x40, Don't fragment
  - ...0 0000 0000 0000 = Fragment Offset: 0
  - Time to Live: 64
  - Protocol: UDP (17)
  - Header Checksum: 0xf44b [validation disabled]
  - [Header checksum status: Unverified]
  - Source Address: 10.0.2.6
  - Destination Address: 5.135.215.43
- User Datagram Protocol, Src Port: 7078, Dst Port: 12134
- Real-Time Transport Protocol

| *enp0s3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |          |              |          |        |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|--------------|----------|--------|-----------------|
| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |          |              |          |        |                 |
| Apply a display filter ... <Ctrl-/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |          |              |          |        |                 |
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Time        | Source   | Destination  | Protocol | Length | Info            |
| 118                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.183964111 | 10.0.2.6 | 5.135.215.43 | STUN     | 62     | Binding Request |
| ▶ Frame 118: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface enp0s3, id 0<br>▶ Ethernet II, Src: PcsCompu 55:44:07 (08:00:27:55:44:07), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)<br>▶ Destination: 52:55:0a:00:02:01 (52:55:0a:00:02:01)<br>▶ Source: PcsCompu 55:44:07 (08:00:27:55:44:07)<br>Type: IPv4 (0x0800)<br>▶ Internet Protocol Version 4, Src: 10.0.2.6, Dst: 5.135.215.43<br>0100 .... = Version: 4<br>.... 0101 = Header Length: 20 bytes (5)<br>▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)<br>Total Length: 48<br>Identification: 0x5daf (23983)<br>▶ Flags: 0x40, Don't fragment<br>...0 0000 0000 0000 = Fragment Offset: 0<br>Time to Live: 64<br>Protocol: UDP (17)<br>Header Checksum: 0xf455 [validation disabled]<br>[Header checksum status: Unverified]<br>Source Address: 10.0.2.6<br>Destination Address: 5.135.215.43<br>▶ User Datagram Protocol, Src Port: 7078, Dst Port: 12134<br>▶ Session Traversal Utilities for NAT |             |          |              |          |        |                 |

| *enp0s3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |          |              |          |        |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|--------------|----------|--------|---------------|
| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |          |              |          |        |               |
| Apply a display filter ... <Ctrl-/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |          |              |          |        |               |
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Time        | Source   | Destination  | Protocol | Length | Info          |
| 191                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3.573322546 | 10.0.2.6 | 5.135.215.43 | RTCP     | 320    | Sender Report |
| ▶ Frame 191: 320 bytes on wire (2560 bits), 320 bytes captured (2560 bits) on interface enp0s3, id 0<br>▶ Ethernet II, Src: PcsCompu 55:44:07 (08:00:27:55:44:07), Dst: 52:55:0a:00:02:01 (52:55:0a:00:02:01)<br>▶ Destination: 52:55:0a:00:02:01 (52:55:0a:00:02:01)<br>▶ Source: PcsCompu 55:44:07 (08:00:27:55:44:07)<br>Type: IPv4 (0x0800)<br>▶ Internet Protocol Version 4, Src: 10.0.2.6, Dst: 5.135.215.43<br>0100 .... = Version: 4<br>.... 0101 = Header Length: 20 bytes (5)<br>▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)<br>Total Length: 306<br>Identification: 0x5e7c (24188)<br>▶ Flags: 0x40, Don't fragment<br>...0 0000 0000 0000 = Fragment Offset: 0<br>Time to Live: 64<br>Protocol: UDP (17)<br>Header Checksum: 0xf286 [validation disabled]<br>[Header checksum status: Unverified]<br>Source Address: 10.0.2.6<br>Destination Address: 5.135.215.43<br>▶ User Datagram Protocol, Src Port: 7078, Dst Port: 12134<br>▶ Real-time Transport Control Protocol (Sender Report) |             |          |              |          |        |               |

## Conclusion

In this lab, VoIP interception and traffic analysis were successfully carried out in a controlled virtual environment. A call was initiated using Linphone, and the network traffic generated during the communication was captured using Wireshark. The captured packets revealed important VoIP signaling and media transmission details, including protocol types, IP addresses, and session information.

The lab demonstrated how VoIP communication works at the packet level and highlighted potential security risks when encryption is not enabled. It also emphasized the importance of secure VoIP configurations to prevent unauthorized interception.

Overall, the experiment enhanced practical knowledge of VoIP technology, packet analysis, and network security monitoring techniques.